HYBRID POWER SOURCE IN FISH FARMING WITH HELP OF IOT TECHNOLOGY

A PROJECT REPORT

Submitted by

RAGAVI R	**E20ECR058**
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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

E.G.S. PILLAY ENGINEERING COLLEGE (AUTONOMOUS)

NAGAPATTINAM-611002

ANNA UNIVERSITY: CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this project report **"HYBRID POWER SOURCE IN FISH FARMING WITH HELP OF IOT TECHNOLOGY"** is the bonafide work of **"RAGAVI R (E20ECR058), RAMYA A (E20ECR061), SOWNDHARYA G (E20ECR081), SUNMATHY S (E20ECR085)"** who carried out the project work under my supervision.

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INTERNAL EXAMINER EXTERNAL EXAMINER

DECLARATION

We submit this project work entitled **"HYBRID POWER SOURCE IN FISH FARMING WITH HELP OF IOT TEHNOLOGY"** In partial fulfil of the requirements for the award of degree of **"BACHELOR OF ENGINEERING".** We declare that it was carried out independently by us under the guidance of **Dr. S. SENTHILKUMAR M.Tech.,Ph.D.,** Assistant professor of electronics and communication engineering department.

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ACKNOWLEDGEMENT

Indeed, it is our responsibility to express our heartfelt gratitude to our late founder and Chairman, Chevalier **Dr. G.S. Pillay**. We aim to honor our chairman through our obedience. Our Honorable Chairman **Mrs. S. JOTHIMANI AMMAL G.S. PILLAY** a good hearted, vital of our institution.

We would like to thank our honorable Secretary **Shri. S. SENTHIL KUMAR** and our honorable Joint Secretary **Shri. S. SANKAR GANESH,** for having a great dream to serve for the nation by educating the rural community with the help of their institution.

We also express our sincere thanks to our Principal **Dr. S. RAMABALAN,** M.E., Ph.D., for providing us with the necessary facilities to carry out this project work and our Academic director **Dr. S. PALANI MURUGAN**, M.E., Ph.D., for showing us the right path.

We are very much honored and would like to profound our deep grateful to our Head of the department **Dr. M. MALATHI**, M.E., Ph.D., for her expert, sincere and valuable guidance to us. We all are must to express our sincere thanks to our project coordinators **Dr. M. NUTHAL SRINIVASAN,** M.E., Ph.D., and **Dr. M. IRSHAD AHAMED,** M.Tech., Ph.D., for their full support and motivation given to us.

We would like to express our sincere thanks to our project guide **Dr.S. SENTHILKUMAR,** M.Tech.,Ph.D., for his wide knowledge, serious research attitude and enthusiasm in work deeply impressed and made us to complete this project to the at most satisfaction of our superiors in a stipulated period.

We express our hearty render and thanks to all the staff members and technical assistants of the **Department of Electronics and Communication Engineering** for their timely assistance. Our humble gratitude and thanks to our parents and friends who have supported and help us to complete the project work.

ABSTRACT

Fish farming is a challenging job for producing fish without any hazard. By maintaining the water quality of the pond can safe fish production. This will discuss the development of a real-time water quality system to monitor fish farm. The water quality monitoring system consist of a dissolved oxygen sensor, pH sensor and water temperature sensor using Arduino Nano as the microcontroller, integrated with Internet of Things (IoT) platform is set up to enable the real-time remote water monitoring system. The system is designed to efficiently monitor the water quality in fresh-water fish farm, specifically the water parameters suitable for the fish. Instead of testing the system in the real fish farm, it is tested using a water that is simulated to the water quality for the fish farm. if the water was above and below the fixed parameter it give us an alert and water exchange mechanism get started by remove the dirty water from the pond through the solenoid valve and provide the fresh water into the pond. By measuring the dissolved oxygen level if is any deviation occurs water circulation mechanism should be established and circulate the water for increasing the oxygen level by using the padding wheel which all these done automatically. It also done the automatic food feeding at the periodic interval time without the manual involvement sometimes it may cause an manual error such as underfeeding and overfeeding which may affect the fish health to reduce such stuffs we use this automatic food feeding system.

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CHAPTER 1
INTRODUCTION

Hybrid power source in fish farming with help of IOT technology using a combination of renewable energy sources like solar or wind power, along with traditional power sources, to create a more sustainable and efficient system. With IoT, sensors and devices can be used to monitor and control various aspects of fish farming, such as water quality, oxygen levels, and feeding systems. This helps optimize the farming process and reduce energy consumption. It's a great example of how technology can benefit the environment and improve productivity in fish farming. IoT technology can be used to monitor and control the feeding process. Imagine a system where sensors are placed in the fish tanks to measure factors like water temperature, pH levels, and oxygen levels. These sensors can send real-time data to a central control unit, which can then analyse the information and determine the optimal feeding schedule for the fish. The control unit can also be connected to an automated feeding system that dispenses the right amount of food at the right time, ensuring that the fish are well-fed and healthy. This not only improves the efficiency of the feeding process but also reduces waste and ensures that the fish receive the right nutrients. the need for sustainable and efficient power solutions is crucial. That's where hybrid power sources come into play. By combining renewable energy sources like solar or wind power with traditional power sources, fish farmers can create a more environmentally friendly and cost-effective system. With the integration of IoT technology, fish farmers can take their operations to the next level. IoT allows for the seamless connection of various devices and sensors, enabling real-time monitoring and control of important factors in fish farming, such as water quality, oxygen levels, and feeding systems. By leveraging IoT, fish farmers can optimize their operations, reduce energy consumption, and improve the overall health and

growth of their fish. It's an innovative approach that not only benefits the environment but also enhances productivity in fish farming.

The demand for sustainable and efficient power solutions has become increasingly important. That's where hybrid power sources come in. These systems combine renewable energy sources, such as solar or wind power, with traditional power sources like electricity or diesel generators. By integrating these different power sources, fish farmers can achieve a more environmentally friendly and cost-effective approach to powering their operations. But here's where it gets even more exciting! With the integration of IoT technology, fish farmers can unlock a whole new level of control and optimization. The Internet of Things allows for seamless connectivity between devices and sensors, enabling real-time monitoring and management of critical aspects in fish farming. Imagine the sensors placed in fish tanks can measure parameters like water quality, temperature, and oxygen levels. These sensors can then send data to a central control unit or a cloud-based platform through wireless communication. The control unit analyses the data and provides valuable insights to fish farmers, helping them make informed decisions about feeding schedules, water circulation, and other environmental factors. Not only that, but fish farmers can remotely access the system through mobile or web interfaces. This means they can monitor their fish farm, receive alerts, and make necessary adjustments from anywhere, anytime. By leveraging hybrid power sources and IoT technology, fish farmers can optimize energy consumption, reduce operational costs, and improve the overall health and growth of their fish. It's a win-win situation for both the environment and productivity in fish farming. So, picture a future where fish farms harness the power of renewable energy, combined with smart IoT systems, to create sustainable and efficient fish farming practices. It's an exciting prospect that holds great potential for the future of aquaculture.[1]

CHAPTER 2
LITERATURE REVIEW

Detection of water quality by turbidity sensor and PH level sensor – 2019 – Mandalay university, Myanmar.

The system is detected the turbidity and PH level of water. The system is controlled by Arduino Uno board between two input sensors and the output liquid crystal display. The turbidity sensor is determined the water quality by using photometric technique. The PH level sensor is tested the acidity and alkalinity of a solution. The results are displayed on Liquid crystal display screen. The program is written by Arduino programming language. Basically, the PH value is a detection how water is hard or soft. The pH of pure water is 7. Liquid with a pH lower than 7 is acidic, and with a pH greater than 7 is basic. Alkalinity is a measure of the ability of the water to resist a change in pH that would tend to make the water more acidic. The measurement of alkalinity and pH is needed to control the corrosiveness of the water. Turbidity is the cloudiness of a fluid caused by large numbers of individual particles that are generally invisible to the uncovered eye, similar to smoke in air. The measurement of turbidity is a main test of water quality.

IOT Based water quality measurement system using ESP32 – 2021 – Horizon institute of technology, Thane.

Water pollution is one of the biggest horror for the green globalization. And we all know about Coronavirus Pandemic how dangers it is. In such case water measurement is very important for human life. In order to ensure the safe supply of the drinking water the quality needs to be measure and monitoring. In this paper we present a IOT (Internet Of Things) based water quality measuring system using Arduino. The system consist of various sensors is used to measuring parameters of the water. The parameters such as temperature, PH, turbidity,

sensor of the water can be measured. The measured values from the Different sensors can be processed by the Arduino core controller. These sensors relate to Arduino for the purpose of measurement the parameters of water quality . Finally, the sensor data can be viewed on internet by user using WIFI Module.it had been found that the system works reliably but is reliant on human assistance and susceptible to data inaccuracies. The system however, provides a solid foundation for future expansion works of an equivalent category to elevate the system to being Internet of Things (IoT) friendly And Therefore, this paper aims to investigate the feasibility of implementing an Arduino -based sensor system for water quality monitoring.

Automatic animal food feeding system – 2021 – P.V.P institute of technology, Maharashtra.

India dominates among other countries in terms of agriculture production, and is one of the largest milk-producing country in the world. A huge human population in India relates itself to agriculture and livestock. Because of this, there are a massive number of cattle farms & dairy farms in India. Also, there is large number of poultries present. According to a survey conducted by the national dairy development board in 2019, there are more than 192 million cattle in India and is at least one personal is required for every 10 animals. In gigantic dairy farming the manpower required is also huge as well as the operations are very laborious and time consuming. By the use of automated machines, it is possible to control the number of required man power, and to avoid financial loss. The time required for process is also reduced. The main objective of the project discussed in this paper is to introduce a laboratory model for automation in animal food feeding process using instrumentation and control strategies. It is thought that the implemented model discussed in this paper can also be used with AMS (automatic milking system) in future.

Survey to analyse the parameters and problems in fish farming – 2024 – College of Fisheries University, Tamil Nadu Dr.J.Jayalalithaa Fisheries University (TNJFU), Nagapattinam.

We visited the fish farming pond in the fisheries college to analyse detailed about the fish farming thus may help us to know about the parameters like TDS must be maximum 1000 ppm. we have to monitor or maintain the waste of fish (ammonia) is the very must process to avoid the toxic formation in the pond the ammonia content must be less than 0.1 to 0.2 then Nitrite level must be at 0.1 ppm if we place the fish farm in open space thus nitrate is required the ranges upto 500ppm. Salinity level has to be maintained between 0.5 to 1. From that survey they strictly illustrated not to use chlorine and then pond depth should be minimum 1metre. Food contamination must be 10% of its body weight per day and protein level should be around 24% . Mainly focus on water PH level that should be maintained at 7.4 to 8 then Temperature is to maintained at 18-32 degree Celsius. To maintain the water with the correct PH level is mandatory in case there may any change dolomite and quicklime are used to increase the PH level. carbon-di-oxide is used to decrease the PH level.

Figure 2.1 & 2.2 Visiting the Fisheries Engineering College

CHAPTER 3

TYPES OF FISH FARMING

3.1 CAGE SYSTEM:

Cage system in fish farming method is where fish are raised in netted enclosures in natural bodies of water like lakes or oceans. These cages keep the fish contained while allowing water to flow through. This helps in controlling the fish's environment, protecting them from predators, and making it easier to manage their feeding and growth. It's a popular method because it maximizes space and helps promote sustainable fish farming.

Figure 3.1 cage system

3.2 OPEN NET PEN SYSTEM:

The open net pen system in fish farming. It's a method where fish are raised in large, floating net enclosures placed directly in natural bodies of water like lakes, rivers, or coastal areas. These net pens are made of sturdy mesh material and are secured to the seabed or anchored in place. In the open net pen system, the fish are contained within the nets, allowing them to swim and interact with the surrounding water. This system takes advantage of the natural environment while still providing some control over the fish farming process.

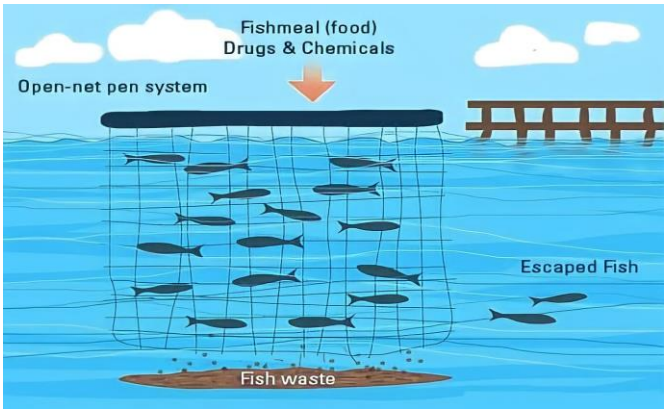


Figure 3.2 Open net pen system

3.3 IRRIGATION DITCH OR POND SYSTEM:

An irrigation ditch is a channel that is used to divert water from a water source, such as a river or a reservoir, to the fish farm. The water flows through the ditch and can be directed into different areas of the farm to provide water for the fish. These ponds are usually dug into the ground and lined with a waterproof material to prevent water from seeping into the soil. The ponds are filled with water and stocked with fish. The advantage of a pond system is that it allows for better control over water quality, temperature, and feeding practices. Both irrigation ditch and pond systems have their own benefits and considerations. The choice depends on factors like available resources, farm size, and the type of fish being raised. It's important to consider factors like water source, land availability, and local regulations when deciding which system to use.

Figure 3.3 Irrigation ditch or pond system

CHAPTER 4

PROPOSED SYSTEM

4.1 EXISTING SYSTEM:

In the existing model of the fish farm management system we often see that the dedicated individuals feeding the fish through out the day which may cause an human error such as overfeeding and underfeeding which can affect the health and fish growth . people has to regularly observe the fish and keeping the close eye on their behaviour and health due to the inconsistent monitoring this could lead to delayed responses to changes in water quality or fish health maintaining tasks like cleaning the tank and debris removal are performed by hand where the farm requires a dedicated workforce to perform various tasks this can result in higher labour costs . Manual management may lack the ability to collect the detailed data on parameters such as feeding patterns ,growth rates ,and water quality which can be the hinder optimization and decision making.

4.2 PROPOSED SYSTEM:

In fish farming, the demand for sustainable and efficient power solutions has become increasingly important. That's where hybrid power sources come in. These systems combine renewable energy sources, such as solar or wind power, with traditional power sources like electricity or diesel generators. By integrating these different power sources, fish farmers can achieve a more environmentally friendly and cost-effective approach to powering their operations. With the integration of IoT technology, fish farmers can unlock a whole new level of control and optimization. The Internet of Things allows for seamless connectivity between devices and sensors, enabling real-time monitoring and management of critical aspects in fish farming. sensors placed in fish ponds can measure parameters like water quality, temperature, and oxygen levels, salinity, PH and toxic subtances. These sensors can then send data to a central control unit or a

cloud-based platform through wireless communication. The control unit analyses the data and provides valuable insights to fish farmers, helping them make informed decisions about feeding schedules, water circulation, and other environmental factors. Not only that, but fish farmers can remotely access the system through mobile or web interfaces. This means they can monitor their fish farm, receive alerts, and make necessary adjustments from anywhere, anytime. By leveraging hybrid power sources and IoT technology, fish farmers can optimize energy consumption, reduce operational costs, and improve the overall health and growth of their fish.

4.3 BLOCK DIAGRAM:

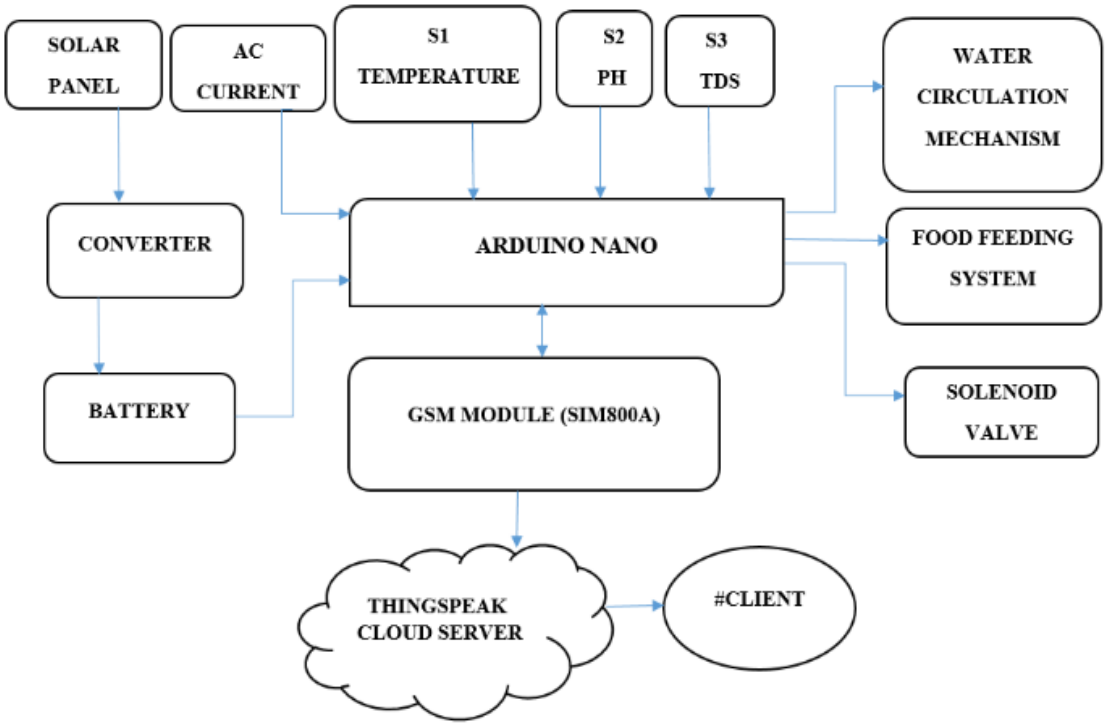


Figure 4.1 Block diagram

4.4 CIRCUIT DIAGRAM:

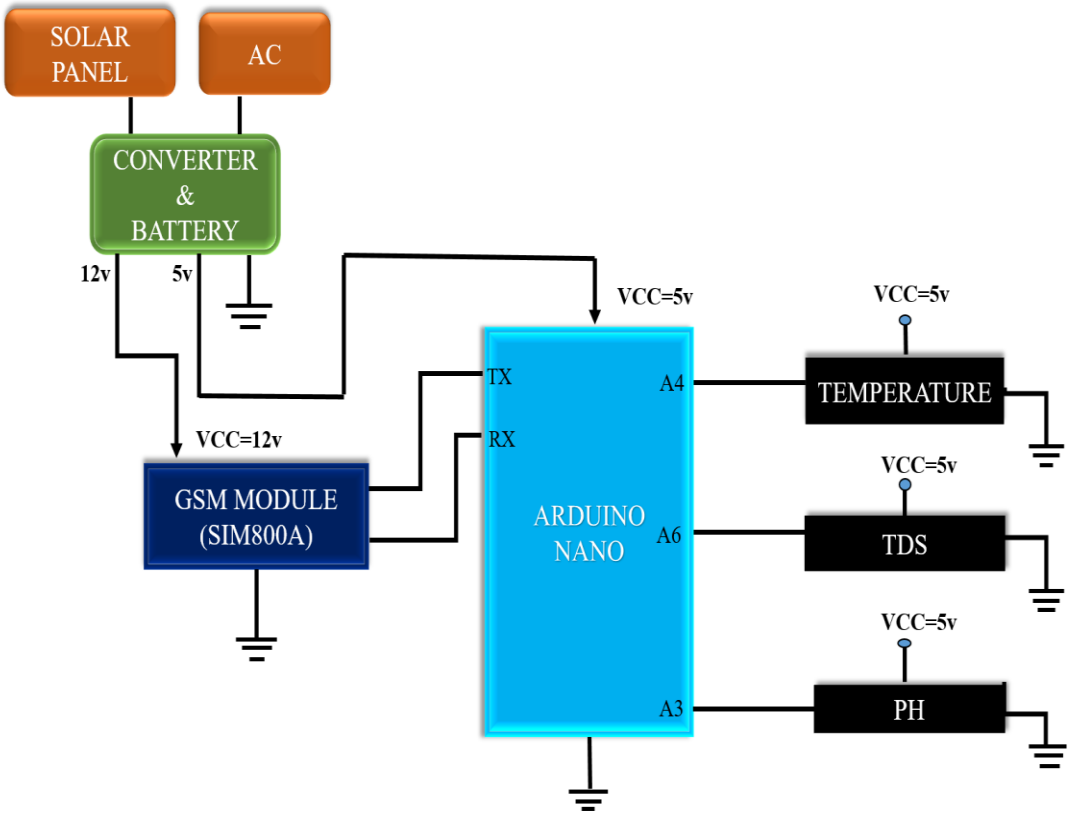


Figure 4.2 Circuit diagram

CHAPTER 5

HARDWARE

5.1 HARDWARE DESCRIPTION:

- GSM Module (SIM800A)
- Temperature Sensor (DS18B20)
- PH Sensor
- TDS Sensor
- Pump with Motor
- Solenoid Valve
- Solar Panel (100w)
- Battery (12v)
- Relay Module
- Buck Boost converter
- Arduino Nano

5.2 GSM MODULE (SIM800A):

The SIM800A is a quad-band GSM/GPRS module that operates on frequencies GSM 850MHz, EGSM 900MHz, DCS 1800MHz, and PCS 1900MHz. It supports both 2G and 2.5G networks, making it compatible with many cellular networks worldwide. The module has various features such as making and receiving calls, sending and receiving SMS messages, and accessing the internet using GPRS. It also supports features like caller ID, call waiting, call forwarding, and USSD (Unstructured Supplementary Service Data) commands. To control the SIM800A module, you can use AT commands. These commands are sent to the module via a serial interface, such as UART, using a microcontroller like Arduino. The module responds to these commands with the desired actions or information.[2]

Figure 5.1 GSM Module (SIM800A)

5.3 TEMPERATURE SENSOR:

The DS18B20 Waterproof Temperature Sensor Probe and experience unmatched performance. This exceptional probe boasts a robust 2-inch steel probe and a 2-meter cable, ensuring you can access the most hard-to-reach places with ease. With its waterproof design, you can use it in any environment without worrying about damage or corrosion, making it perfect for use in industrial settings, greenhouses, aqua ponics systems, and more. But this probe is more than just tough and durable - it's also incredibly precise. Using its high-precision digital thermometer, the DS18B20 can measure temperatures ranging from -55C to +125C with an accuracy of +-0.5C. Installation is effortless with its straightforward 3-wire interface, allowing you to connect and use it with ease. It even communicates over a 1-Wire bus, meaning you can use multiple sensors on the same bus without needing an individual pin for each one. The DS18B20 Waterproof Temperature Sensor Probe is also highly reliable, ensuring that you always get the most accurate temperature readings possible. Plus, its customizable resolution feature allows you to choose between 9, 10, 11, or 12-bit temperature data. Whether you're an engineer, a researcher, or simply looking to monitor the temperature of your pool or hot tub, the DS18B20 Waterproof Temperature Sensor Probe is the perfect solution. [3]

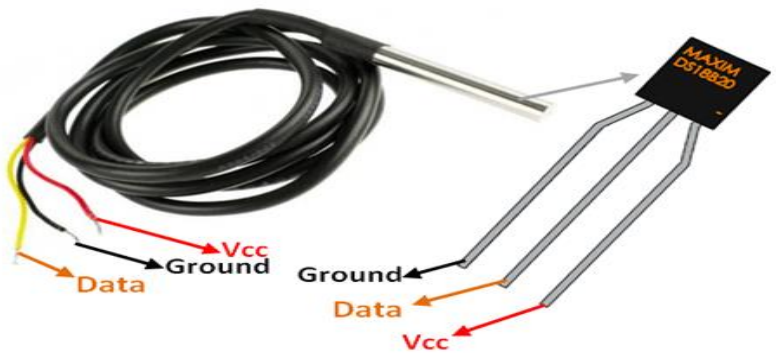


Figure 5.2 Temperature sensor

5.4 PH SENSOR:

PH sensor is one of the most important tools for measuring pH and is commonly used in water quality monitoring. This type of sensor is capable of measuring alkalinity and acidity in water and other solutions. When used properly, pH sensors can ensure the safety and quality of products and processes that occur in wastewater or manufacturing plants. When specializing in water pre-treatment for manufacturing purposes, this type of water quality management allows you to remove impurities from influent water before using it for a variety of different manufacturing processes. With the removal of these impurities, operating efficiency will be improved, heating and cooling efficiency will be higher, and you will be able to maintain compliance with the necessary water quality standards more effectively.[4]

Figure 5.3 PH Sensor

5.5 TDS SENSOR:

TDS is an abbreviation for Total Dissolved Solids in a liquid, including organic and inorganic substances in a molecular, ionic, or micro-granular suspended form. TDS is generally expressed in parts per million (ppm) or as milligrams per litre (mg/L). TDS is directly related to the quality of water will have a TDS between 0 and 10, whereas tap water will vary between 20 and 300, depending on where you live in the world. The materials that constitute dissolved solids in water include materials such as minerals, salts, anionic and cationic substances. If the water contains no soluble materials and is pure, it will not conduct a charge and will, therefore, have a 0 ppm figure. Conversely, if the water is full of dissolved materials, it will conduct a charge, with the resulting ppm figure being proportional to the number of dissolved solids. This is because all dissolved solids have an electrical charge, which allows conduction of electrical charge between the electrode. Fish require a specific TDS and pH similar to the natural environment in which they live. Freshwater fish require less than 400ppm, with some other freshwater fish requiring less. Saltwater fish require TDS readings of between 5000 and 50000ppm.[5]

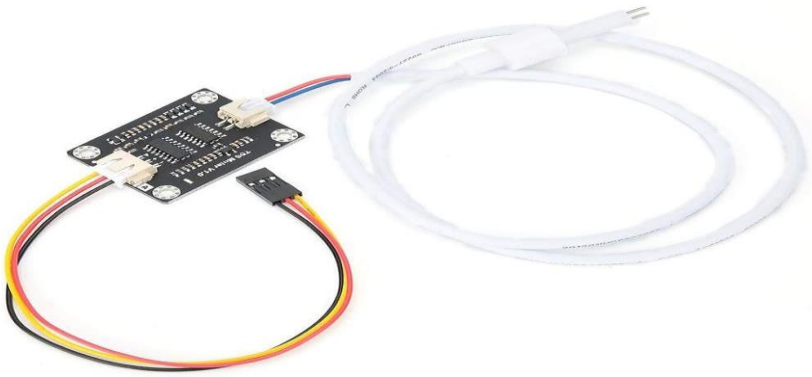


Figure 5.4 TDS Sensor

5.5.1 SPECIFICATION:

1. Input Voltage: 3.3 ~ 5.5V
2. Output Voltage: 0 ~ 2.3V
3. Working Current: 3 ~ 6mA
4. TDS Measurement Range: 0 ~ 1000ppm
5. TDS Measurement Accuracy: ± 10% FS (25 ℃)
6. TDS probe with Number of Needle: 2

5.6 PUMP WITH MOTOR:

12V water pump is powered by dc power supply. People fill its pump chamber with water to remove the air in the water inlet pipe, before the pump motor starts. After the motor is starting, the impeller of the water pump starts to rotate with the rotor under the action of the electromagnetic force generated by the motor, and the water between the blades also moves with the blades. Due to the centrifugal force, water is thrown out from the Center by the impeller, and the water gains kinetic energy, leaving the outer edge of the impeller at a higher speed and entering the volute pump casing. In the volute, water is decelerated due to the bending and lengthening of the flow channel, converting kinetic energy into pressure energy, and flowing into the drainage pipe at a higher pressure. The impeller runs continuously under the drive of the motor. Due to the negative pressure in the pump, under the atmospheric pressure of the horizontal plane, the water is continuously inhaled, and then continuously discharged through the work of the impeller. This is the work of the 12 Volt water pump principle.

5.6.1 GOOD PERFORMANCE:

12v dc water pump has advantages of low repair rate (even zero), small size, high efficiency, low consumption, no pollution, environmental protection, high safety, water proof, static proof, radiation proof, being filled with weak acid, weak alkali, clean water, edible oil, oil, diesel fuel and gasoline.

5.6.2 LOW WORKING NOISE:

The axes of our water pump are made of high quality materials with high precision. The noise of pump work is not more than 35dB, even less than 23dB for low power quiet water pump.

5.6.3 DUTY WORK:

12 volt dc brushless pump can sustain heavy duty work for a day with continuous operation. And its expected life span is at least 30000 hours.

5.6.4 WIDE WORKING CONDITIONS:

12V brushless water pump that made with PPE, PPS material can be applied to indoor and outdoor under temperature 100℃.It is very greatly effective to prevent pump from leak resulting from long time water working by stator and PCB in motor tight encapsulated by peroxide and fully isolating from rotor. It is available for pump to be fixed in water, for the pump water proof. The maximum working liquid temperature is up to 180℃.[6]

Figure 5.5 Pump with Motor

5.7 SOLENOID VALVE:

Solenoid valves are used wherever fluid flow has to be controlled automatically. They are being used to an increasing degree in the most varied types of plants and equipment. The variety of different designs which are available enables a valve to be selected to specifically suit the application in INTERNALLY PILOTED 2-WAY VALVES Internally piloted solenoid valves are fitted with either a 2- or 3-way pilot solenoid valve. A diaphragm or a piston provides the seal for the main valve seat. The operation of such a valve is indicated in Fig. 4. When the pilot valve is closed, the fluid pressure builds up on both sides of the diaphragm via a bleed orifice. As long as there is a pressure differential between the inlet and outlet ports, a shut-off force is available by virtue of the larger effective area on the top of the diaphragm. When the pilot valve is opened, the pressure is relieved from the upper side of the diaphragm. The greater effective net pressure force from below now raises the diaphragm and opens the valve. In general, internally piloted valves require a minimum pressure differential to ensure satisfactory opening and closing. Omega also offers internally piloted valves, designed with a coupled core and diaphragm that operate at zero pressure differential.

5.7.1 TIME DEFINITIONS (VDE0580) RESPONSE TIMES:

The small volumes and relatively high magnetic forces involved with solenoid valves enable rapid response times to be obtained. Valves with various response times are available for special applications. The response time is defined as the time between application of the switching signal and completion of mechanical opening or closing.

5.7.2 ON PERIOD:

The on period is defined as the time between switching the solenoid current on and off.

5.7.3 CYCLE PERIOD:

The total time of the energized and de-energized periods is the cycle period. Preferred cycle period: 2, 5, 10 or 30 minutes.

5.7.4 RELATIVE DUTY CYCLE:

The relative duty cycle (%) is the percentage ratio of the energized period to the total cycle period. Continuous operation (100% duty cycle) is defined as continuous operation until steady-state temperature is reached.

5.7.5 VALVE OPERATION:

The coding for the valve operation always consists of a capital letter. The summary at left details the codes of the various valve operations and indicates the appropriate standard circuit symbols

5.7.6 VISCOSITY:

The technical data is valid for viscosities up to the figure quoted. Higher viscosities are permissible, but in these cases the voltage tolerance range is reduced and the response times are extended.

5.7.7 TEMPERATURE RANGE:

Temperature limits for the fluid medium are always detailed. Various factors, e.g. ambient conditions, cycling, speed, voltage tolerance, installation details, etc., however, influence the temperature performance. therefore, be used only as a general guide.[7]

Figure 5.6 Solenoid Valve

5.8 SOLAR PANEL:

A 100-watt solar panel is one of the most cost-efficient options for running a humble household or a small commercial space. Solar energy generated on-site by your panels has incredible financial and environmental advantages.

5.8.1 RUNS ON SMALL DEVICES:

A panel of 100-watt capacity can provide enough energy to power up your day-to-day appliances for domestic and commercial use. These include lighting systems, mobiles, cable boxes, laptops, and electronic devices.

Fits Small Spaces: If space is a concern, a 100-watt solar panel kit is worth considering. A single panel is 47×21.3×1.4 inches in dimensions, which is relatively small. The module can be installed on the balcony or roof in urban and rural homes.

5.8.2 PORTABILITY:

Solar 100-watt panels are lightweight and conveniently portable. If you are often on the road or in outdoor camps, you can securely carry the solar panels and receive the power you need on the go. Simply mount the panels on your vehicle's roof or camping site to charge your small appliances and devices.

5.8.3 EASY TO INSTALL:

Setting up a 100-watt solar panel system is simple and doesn't require extensive technical knowledge as is necessary for setting up higher-capacity solar systems.

5.8.4 AFFORDABLE:

100w solar panel prices in India are more affordable than traditional energy expenses. The upfront solar investment is a one-time cost that secures 20-25 years of zero-cost electricity supply. This panel size is easily available online and in stores at competitive prices.

5.8.5 WORKING OF A 100W SOLAR PANEL:

A single 100w solar panel comprises a cluster of solar photovoltaic cells manufactured from silicon – a semi-conductive material. When the sunlight falls on the panel's surface, its energy or photons interact with silicon and set electrons in motion in the solar cell. An electric field created by positively and negatively charged semiconductors on the panel directs these electrons to flow towards the conductive metal plates. This creates an energy current in DC form. The DC current passes through a series of wires through the solar inverter, which converts the current into AC electricity. AC energy is used to run appliances and electronics. A 100w 12v solar panel can be tied to a local grid or used as an independent power source in remote locations.[8]

Figure 5.7 Solar Panel

5.9 BATTERY:

12V Lithium LFP battery kit designed for extending run time of fishing boat accessories like depth finders, graphs and lights. Multiple batteries can be combined in parallel for even longer run time. Not intended for trolling motors or as an engine cranking battery. Fishing often means spending countless hours on the water tracking down fish, but to accomplish this task you will need a battery that can power your fish finder reliably no matter where your trip takes you. In order to know which battery is the right choice for your fishing needs, it is important to know what options you have and what the pros and cons of each choice are. Making the right decision is imperative, as a fish finder battery is an invaluable asset that can power many devices on your boat, and a strong one can give you peace of mind while you are out on the water.[9]

Figure 5.8 12V Battery

5.10 RELAY MODULE:

The primary function of a relay module is to switch electrical devices or systems on and off. It also serves to isolate control circuits, ensuring that low-power devices, such as microcontrollers, can safely control higher voltages and currents. This capability is particularly beneficial in scenarios where a small control signal from a microcontroller needs to switch higher currents. In essence, a relay module amplifies this control signal, enabling it to manage more substantial electrical loads The operation of a relay module is relatively straightforward. When a control signal is applied, the electromagnet within the relay activates, leading to the closure of the switch contacts. This action allows the current to flow through the circuit. Conversely, when the control signal is removed, the electromagnet deactivates, opening the switch and interrupting the

current flow. Relay modules can be designed with either normally open (NO) or normally closed (NC) switch configurations. In a NO switch setup, the switch remains open when the electromagnet is not activated and closes upon activation. On the other hand, an NC configuration works oppositely. To ensure the relay module's longevity and prevent potential damage, components like flyback diodes are incorporated. These diodes protect against harmful flyback voltages that can arise when the electromagnet is deactivated. Additionally, optocouplers are sometimes used to offer higher levels of isolation between the control and load circuit.

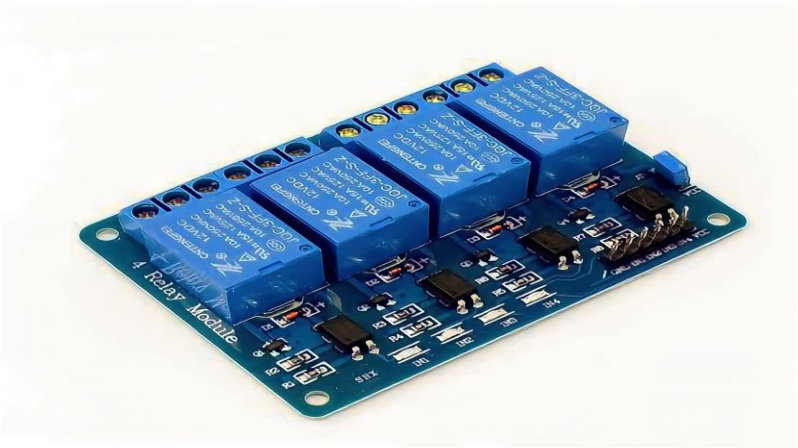


Figure 5.9 Relay Module

5.11 BUCK BOOST CONVERTER:

The buck–boost converter is a type of DC-to-DC converter that has an output voltage magnitude that is either greater than or less than the input voltage magnitude. It is equivalent to a flyback converter using a single inductor instead of a transformer. Two different topologies are called buck–boost converter. Both of them can produce a range of output voltages, ranging from much larger (in absolute magnitude) than the input voltage, down to almost zero.

When a buck (step-down) converter is combined with a boost (step-up) converter, the output voltage is typically of the same polarity of the input, and can be lower or higher than the input. Such a non-inverting buck-boost converter may use a single inductor which is used for both the buck inductor mode and the boost inductor mode, using switches instead of diodes, sometimes called a "four-switch buck-boost converter", it may use multiple inductors but only a single switch as in the SEPIC and Cuk topologies. A buck-boost converter is commonly used to create a constant DC output voltage from a variable DC input voltage source.

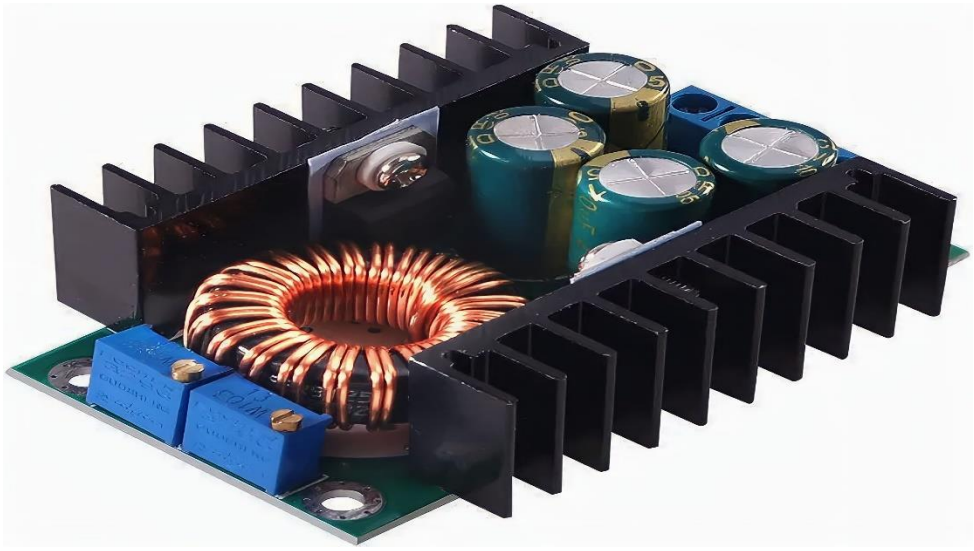


Figure 5.10 Buck Boost Converter

5.12 ARDUINO NANO:

The Arduino Nano is a compact and versatile development board that is based on the ATmega328P microcontroller. It is similar to the Arduino Uno but comes in a smaller form factor, making it ideal for projects where space is limited. The Arduino Nano has 14 digital input/output pins, 8 Analog input pins, and 6 PWM (Pulse Width Modulation) pins. It also has a built-in USB interface for programming and communication with your computer. With the Arduino Nano, you can create a wide range of projects, from simple LED blinking to more

complex robotics and automation systems. It supports the Arduino programming language, which is based on C/C++, and has a large community of users and resources available online.

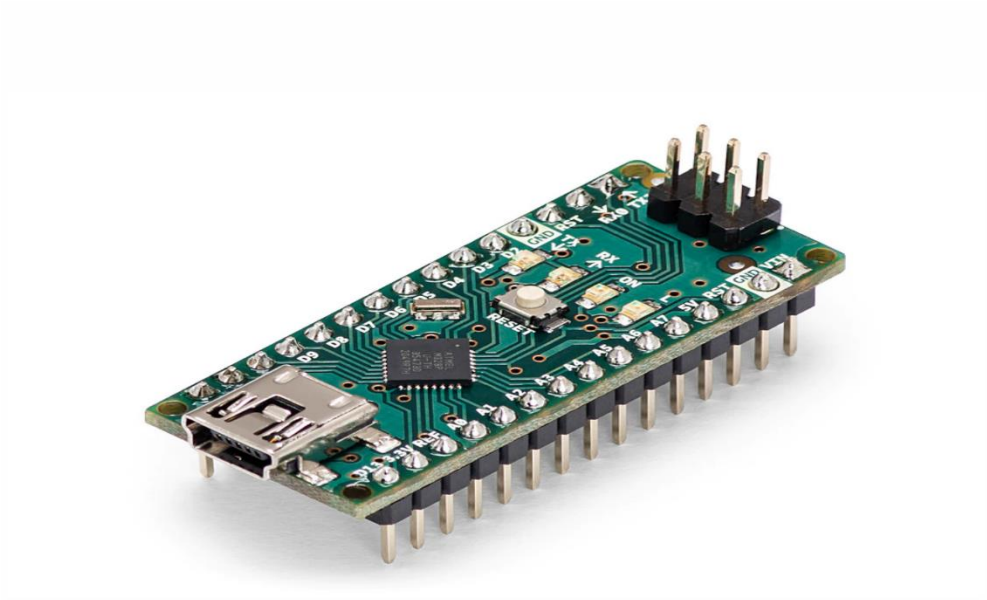


Figure 5.11 Arduino Nano

5.12.1 PIN DIAGRAM OF ARDUINO NANO:

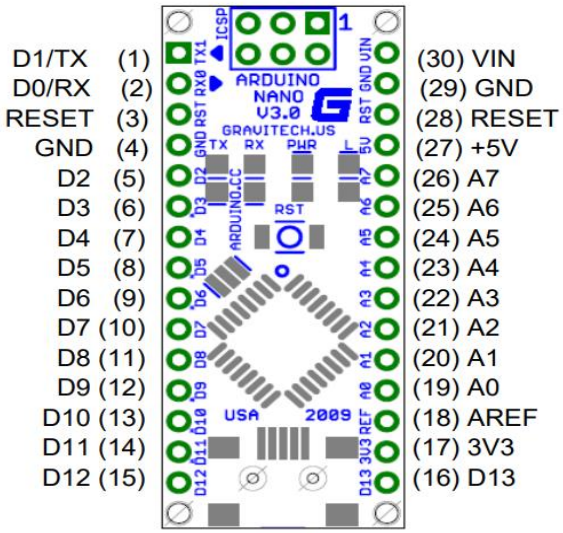


Figure 5.12 Pin Diagram of Arduino Nano

5.12.2 PIN DESCRIPTION:

PIN NO	NAME	TYPE	DESCRIPTION
1-2, 5-16	D0-D13	I/O	Digital input/output port 0 to 13
3, 28	RESET	Input	Reset (active low)
4, 29	GND	PWR	Supply ground
17	3V3	Output	+3.3V output (from FTD)
18	AREF	Input	ADC reference
19-26	A0-A7	Input	Analog input channel 0 to 7
27	+5V	Output or Input	+5V (output from on-broad regulator) +5V (input from external power supply)
30	VIN	PWR	Supply voltage

Table 5.1 Pin Description of Arduino Nano

CHAPTER 6

SOFTWARE DESCRIPTION

6.1 FLOW CHART:

STAGE – 1:

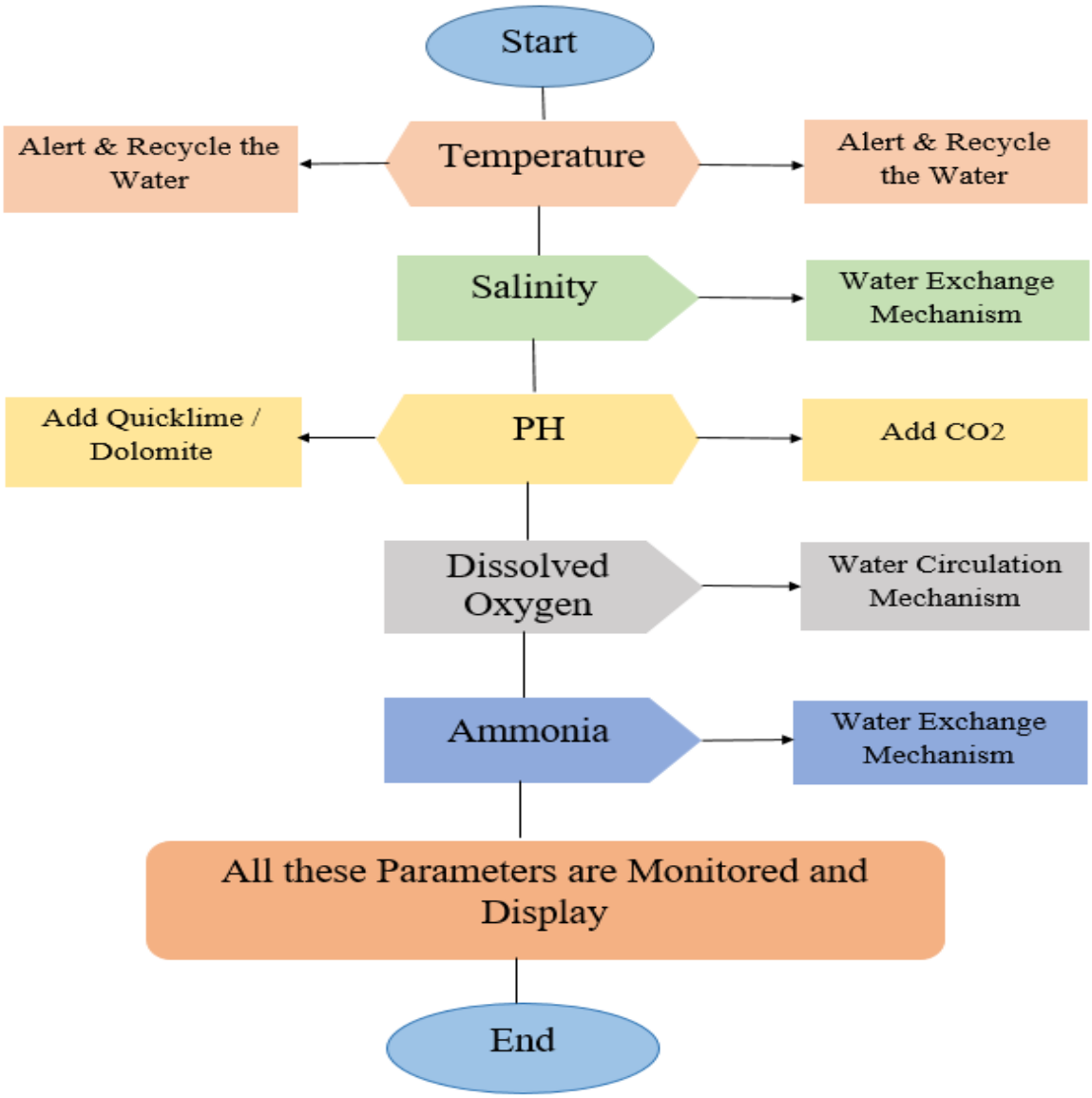


Figure 6.1 Stage 1 of flow chart

STAGE – 2:

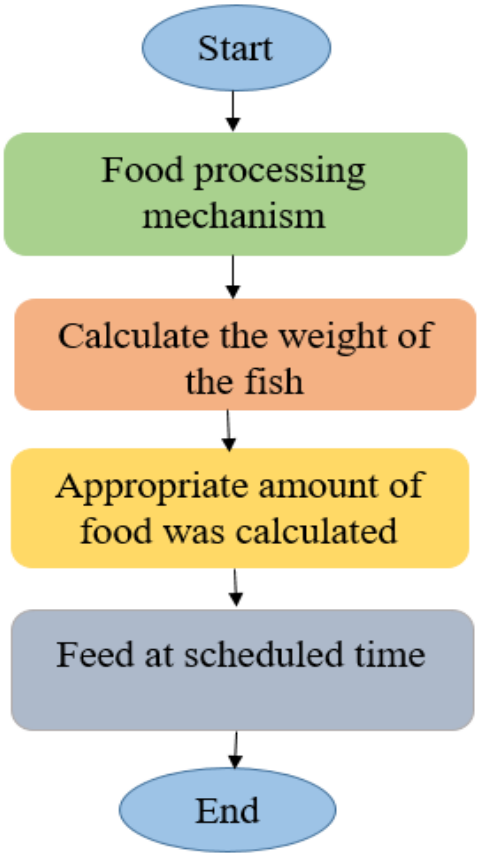


Figure 6.2 Stage 2 of flow chart

6.2 ARDUINO IDE:

The Arduino IDE is like a digital workshop for creating and programming your own electronic projects. It's compatible with various Arduino boards, allowing you to write code in the Arduino programming language, which is based on C/C++. The IDE provides a text editor for writing your code, a compiler to convert it into machine-readable instructions, and a uploader to transfer the code to the Arduino board.

6.2.1 INTERFACING GSM MODULE WITH ARDUINO IDE:

Step 1: Open Arduino IDE.

Step 2: Copy paste the same code into your IDE.

Step 3: Change the phone number and the message you would like to send to the required value.

Step 4: Connect your Arduino to your PC.

Step 5: Select the Board as “Arduino Nano” under Tools.

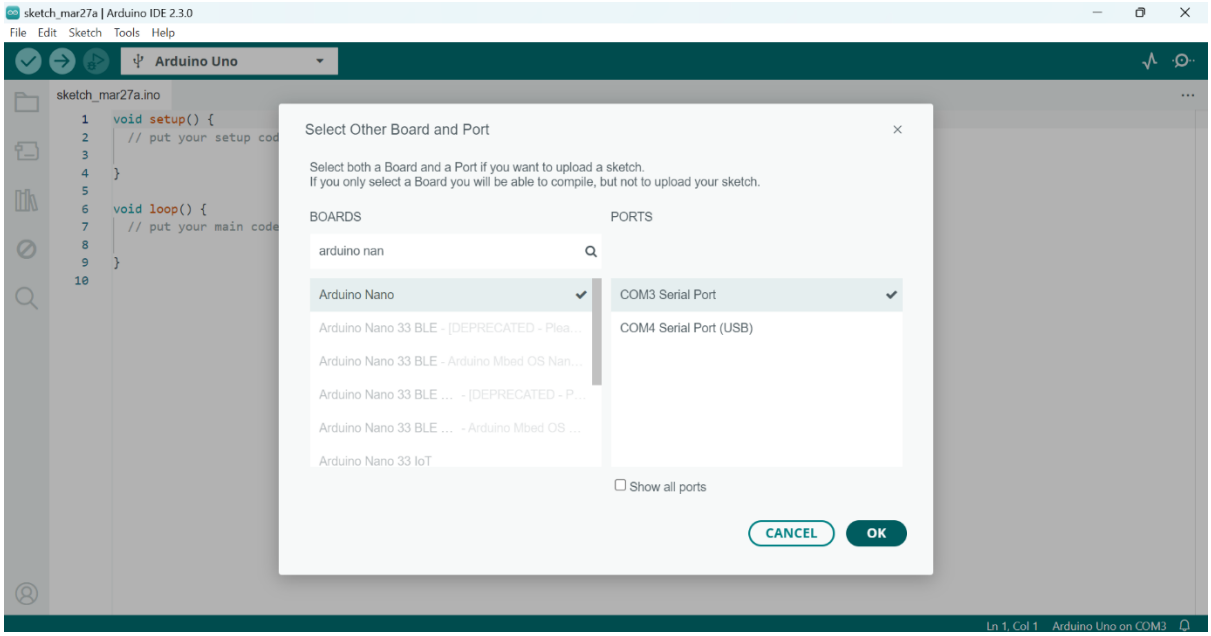


Figure 6.3 Selecting the board and port

Step 6: Select the correct COM port under Tools.

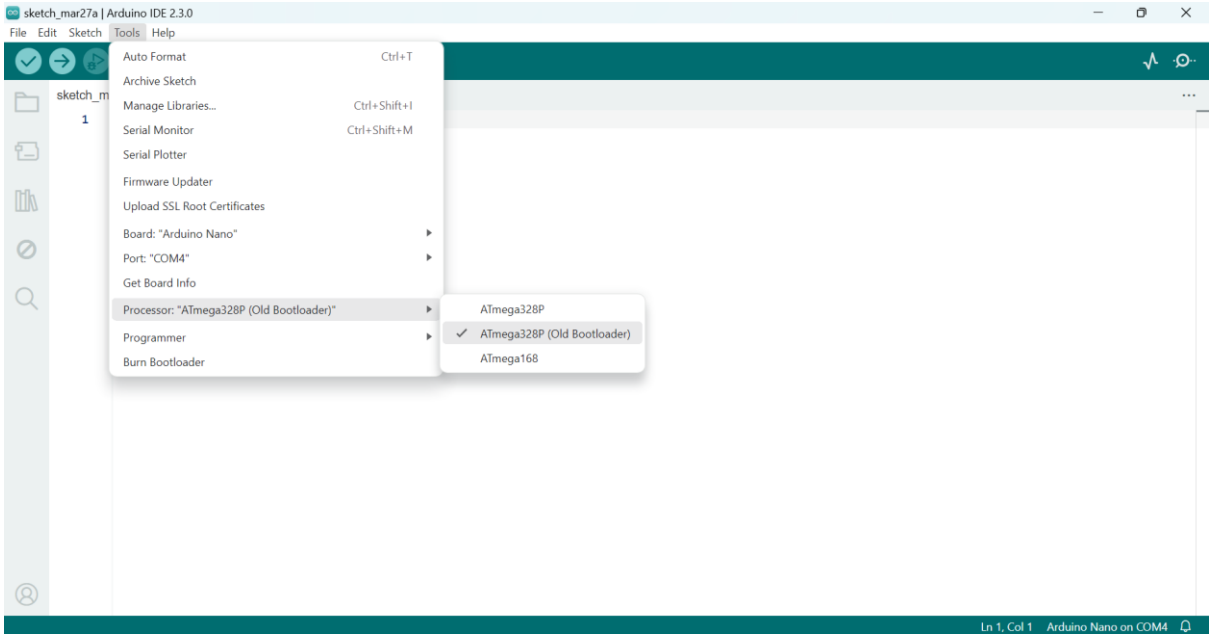


Figure 6.4 Select the old Boot Loader

6.2.2 GSM MODULE (SIM800A) COMMANDS:

COMMAND	DESCRIPTION
OK	Acknowledges execution of a command.
CONNECT	A connection has been established; the DCE is moving from command state to online data state.
RING	The DCE has detected an incoming call signal from network.
NO CARRIER	The connection has been terminated or the attempt to establish a connection failed.

ERROR	Command not recognized, command line maximum length exceeded, parameter vale invalid, or other problem with processing the command line.
NO DIALTONE	No dial tone detected.
BUSY	Engaged (busy) signal detected.
NO ANSWER	"@" (wait for quit answers) dial modifier was used, but remote ringing followed by five seconds of silence was not detected before expiration of the connection timer.

Table 6.1 general commands

6.2.3 Overview of AT commands:

COMMAND	DESCRIPTION
AT+CPIN	Enter PIN
AT+CREG	Network registration
AT+CIICR	Brings up the GPRS or CSD call depending on the configuration previously set by the AT+CSTT command.
AT+CIFSR	Obtain the local IP address and MAC address.
AT+CIPSPRT	Used to set whether echo prompt ">" after issuing "AT+CIP SEND" command.
AT+CIPSHUT	It will close the GPRS PDP Context.
AT+CLIP	Calling line identification presentation.
AT+CLIR	Calling line identification restriction.
AT+CPWD	Change password.
AT+CR	Service reporting control.
AT+CRSM	Restricted SIM access.

AT+CCLK	Clock.
AT+CALM	Alert sound mode.
AT+CALS	Alert sound select.

Table 6.2 AT Commands

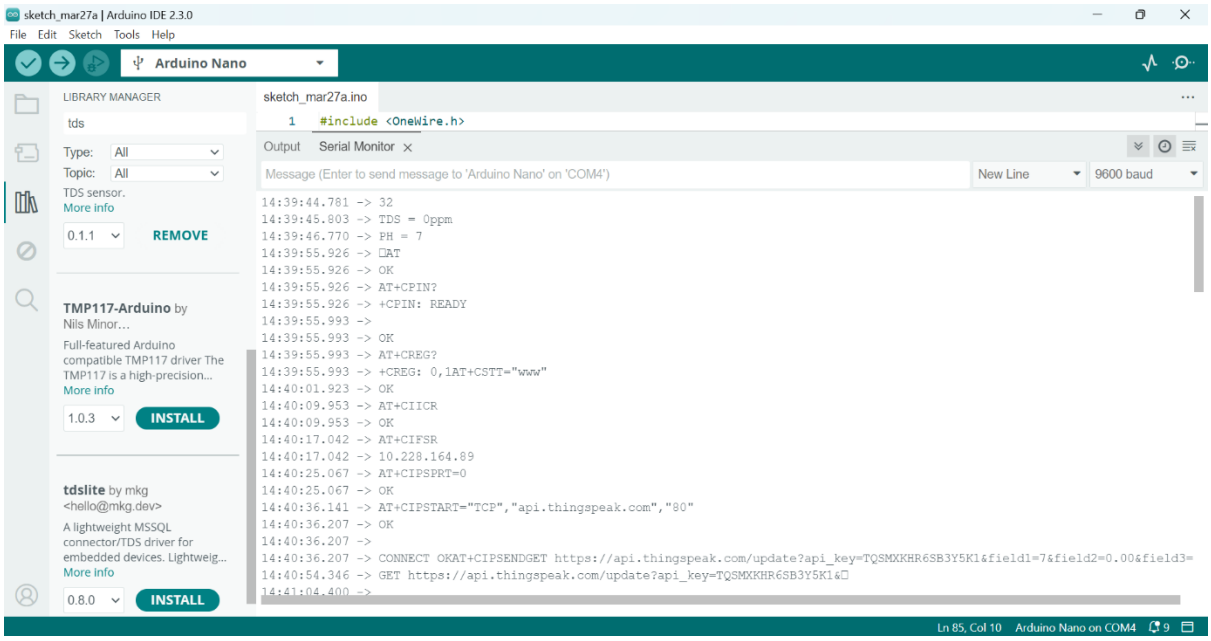


Figure 6.5 GSM Communication protocols

6.3 THINGSPEAK:

ThingSpeak is an Internet of Things (IoT) platform that allow us to collect, analyze, and visualize data from various sensors and devices. It provides an easy-to-use interface and powerful tools to build IoT applications. With ThingSpeak, we can create channels to collect data from sensors, such as temperature, humidity, or motion sensors. We can then visualize the data in real-time using charts, graphs, and gauges. ThingSpeak also supports data analysis and can trigger alerts or actions based on predefined conditions. One of the great features of ThingSpeak is its integration with other platforms and services. We can easily connect our ThingSpeak channels with platforms like MATLAB.[10]

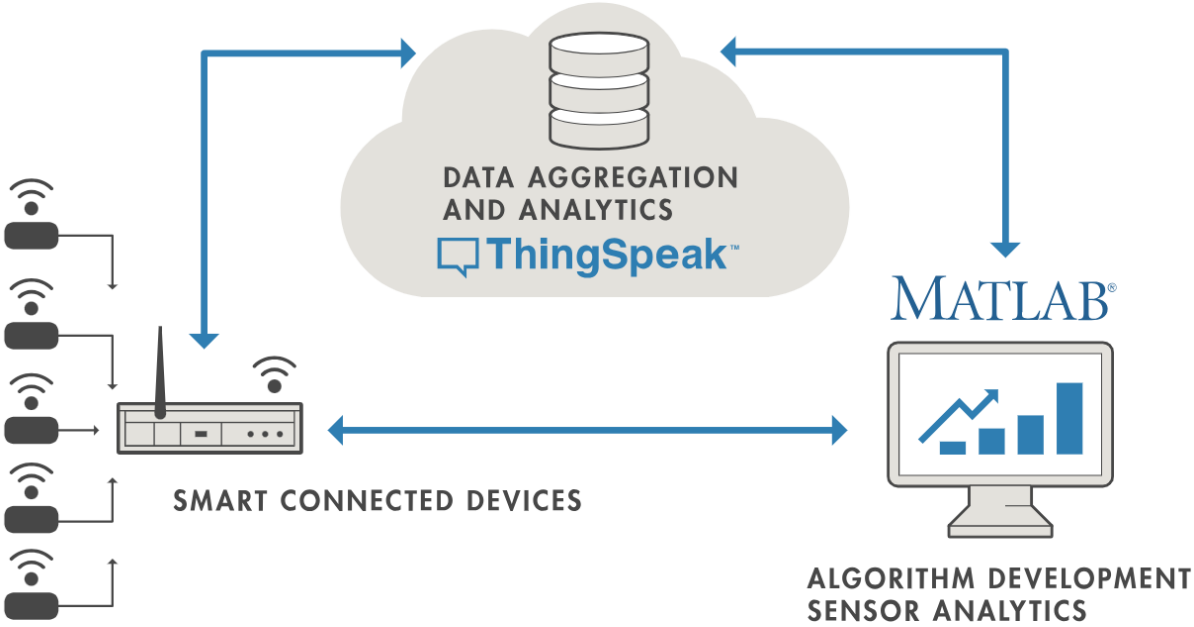


Figure 6.6 ThingSpeak

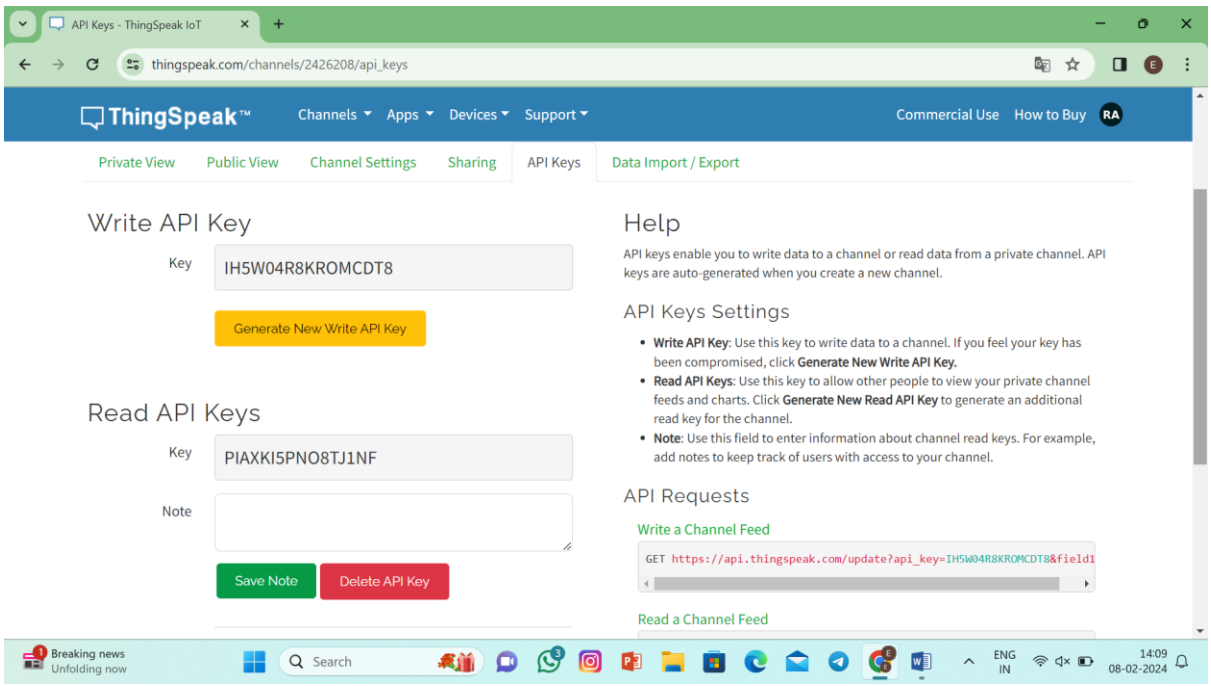


Figure 6.7 ThingSpeak API Key

CHAPTER 7
CONCLUSION

Maintaining the water quality is the major problem that is faced by most of the fish farmers nowadays as it causes a huge fish kills that leads to significant loss of income to fish farmers in. It is impossible to completely avoid fish kill, but in most cases, there are warning signs that the farmers can identify and act on it before it's too late. This is because most of them must check the water quality in their farm manually and it is not very efficient as it needs repeated human intervention to collect and test the sample of the water from the fish farm which cost a lot of time and money. This is where the real-time monitoring is crucial in monitoring the water quality as it will provide the data continuously on the water quality that can be accessed easily by the farmers by using their smartphone or even tablets for efficient water quality management**.** This development has its own advantages and also disadvantages by going through this paper especially based on the results obtained as described here. Overall, the parameters needed in monitoring and maintaining the water quality is investigated which eased the selection of the suitable sensors to be use for the measurement. This development consists of the water quality monitoring sensors connected to the Arduino Nano which then integrated with the IOT platform for data collection. In short, this project is successfully designed to monitor the water quality in the fish farm by collecting the data from the sensors and send it to the cloud server via the Wi-Fi to be accessed by the user using specific IP address for the monitoring purpose through the data displayed on the user interface.

7.1 OUTPUT IMAGES:

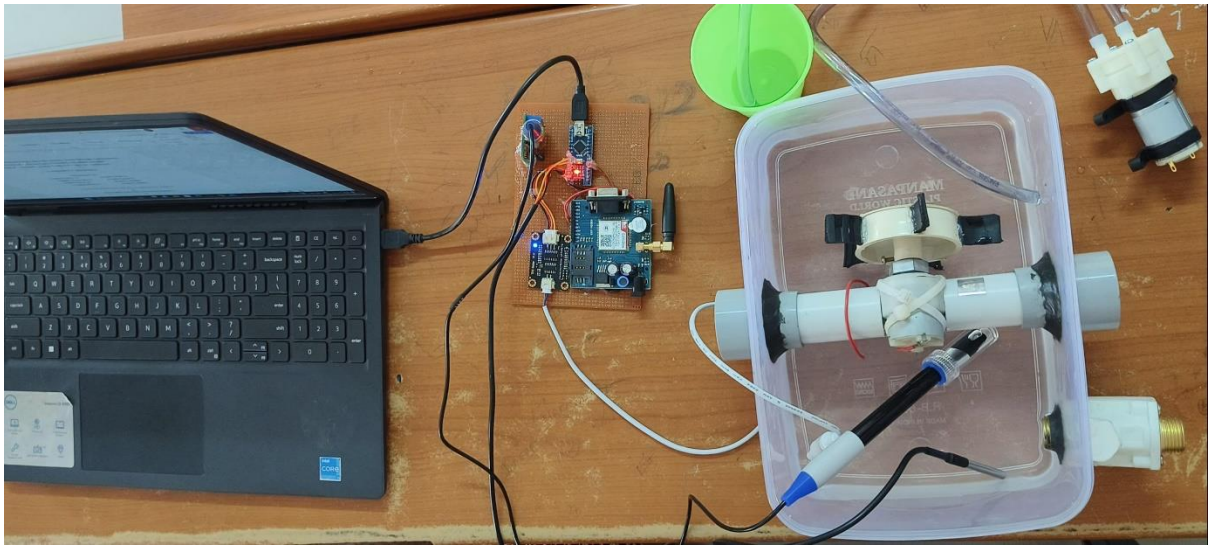


Figure 7.1 Hardware Connections

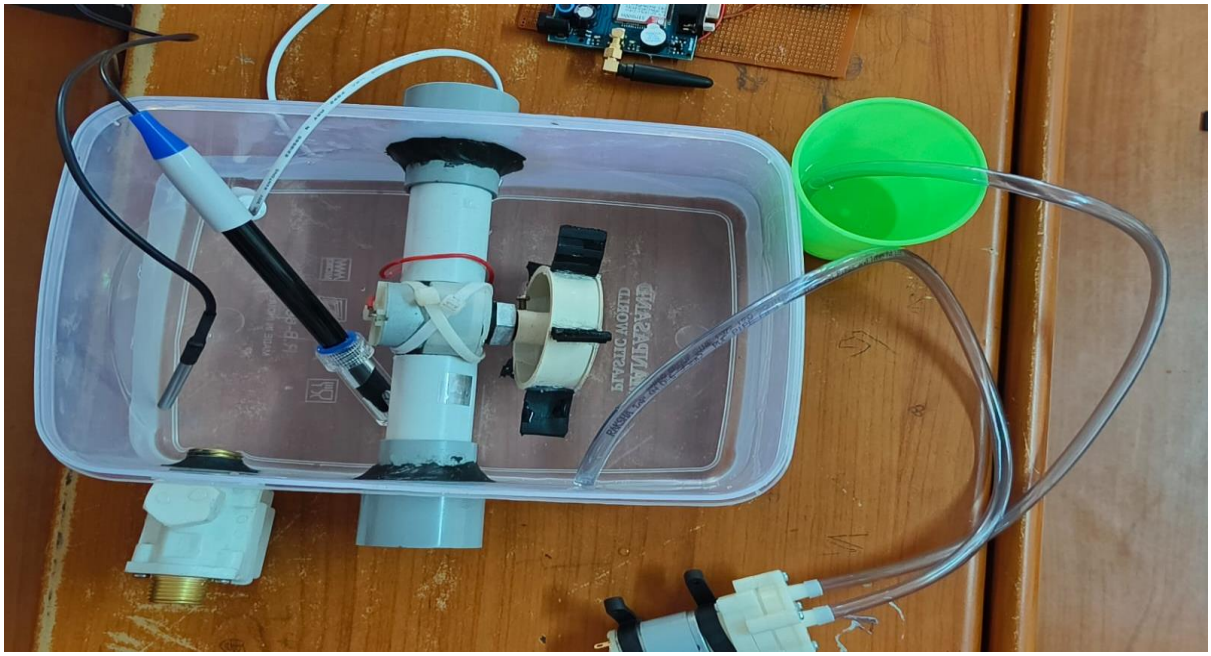


Figure 7.2 Water Circulation Mechanisms

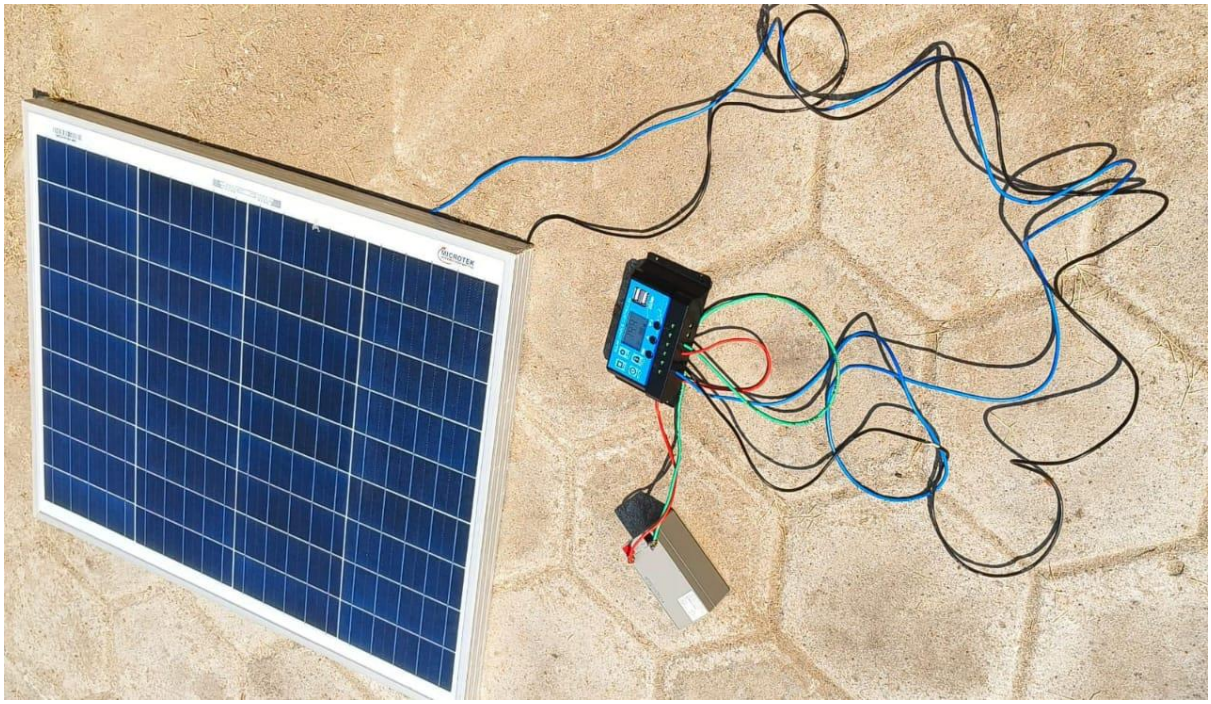


Figure 7.3 Solar Panel Connection

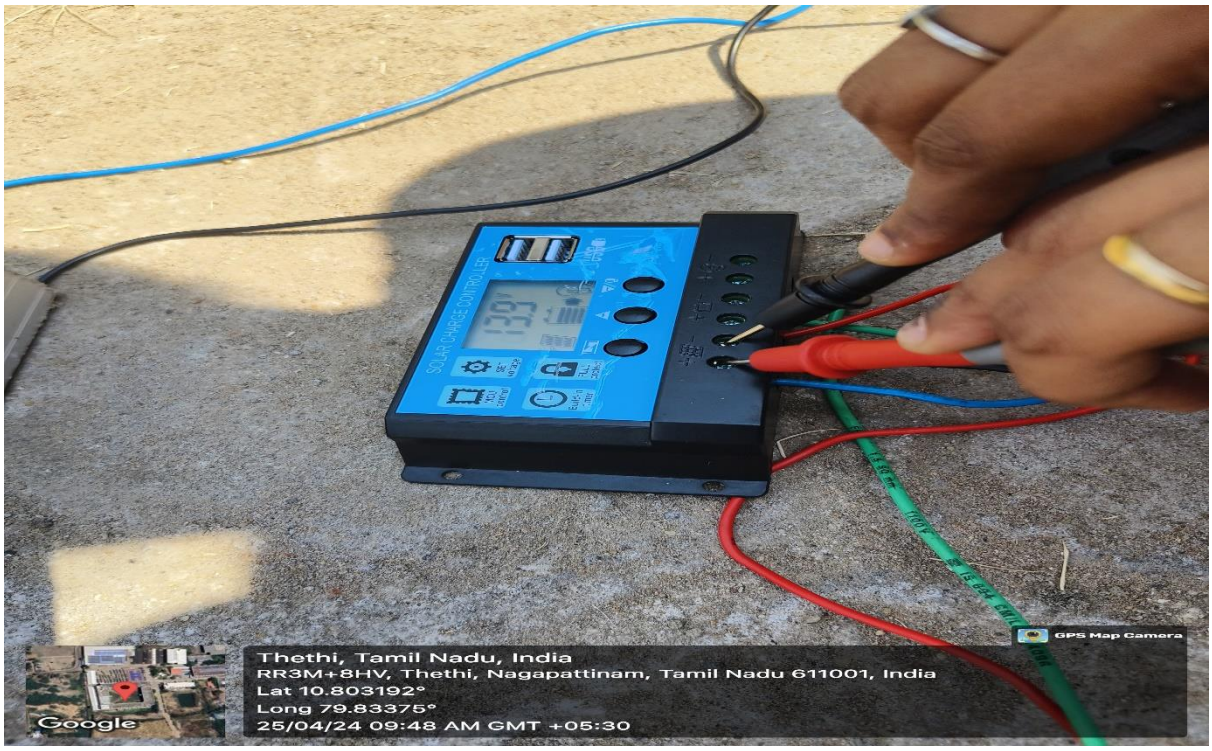


Figure 7.4 Power Generated From the Solar Panel

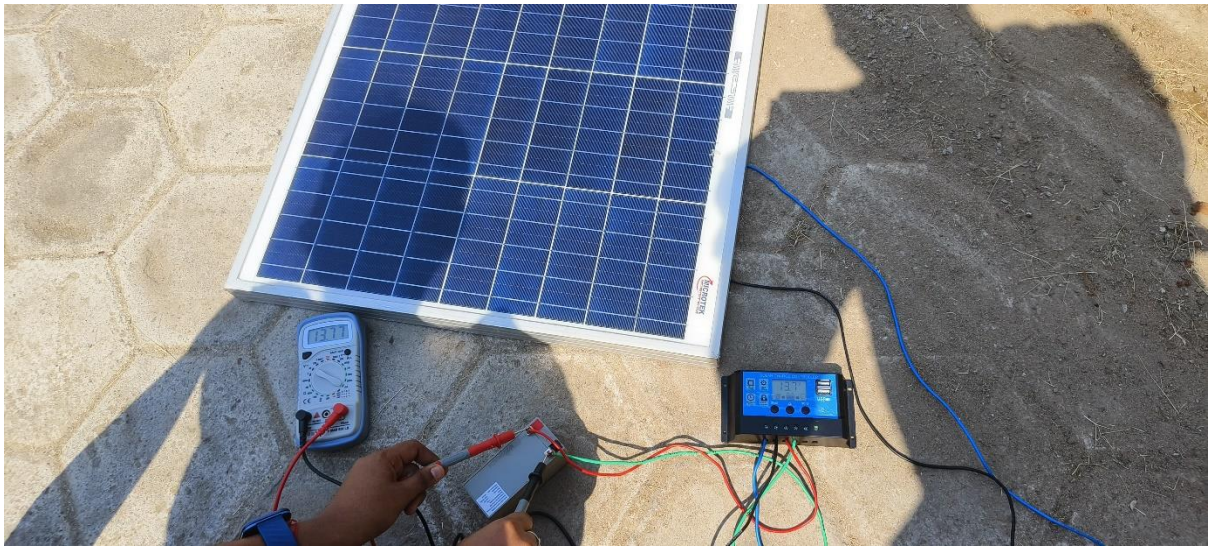


Figure 7.5 Power Stored to the Battery

7. 2 OUTPUT WAVEFORMS:

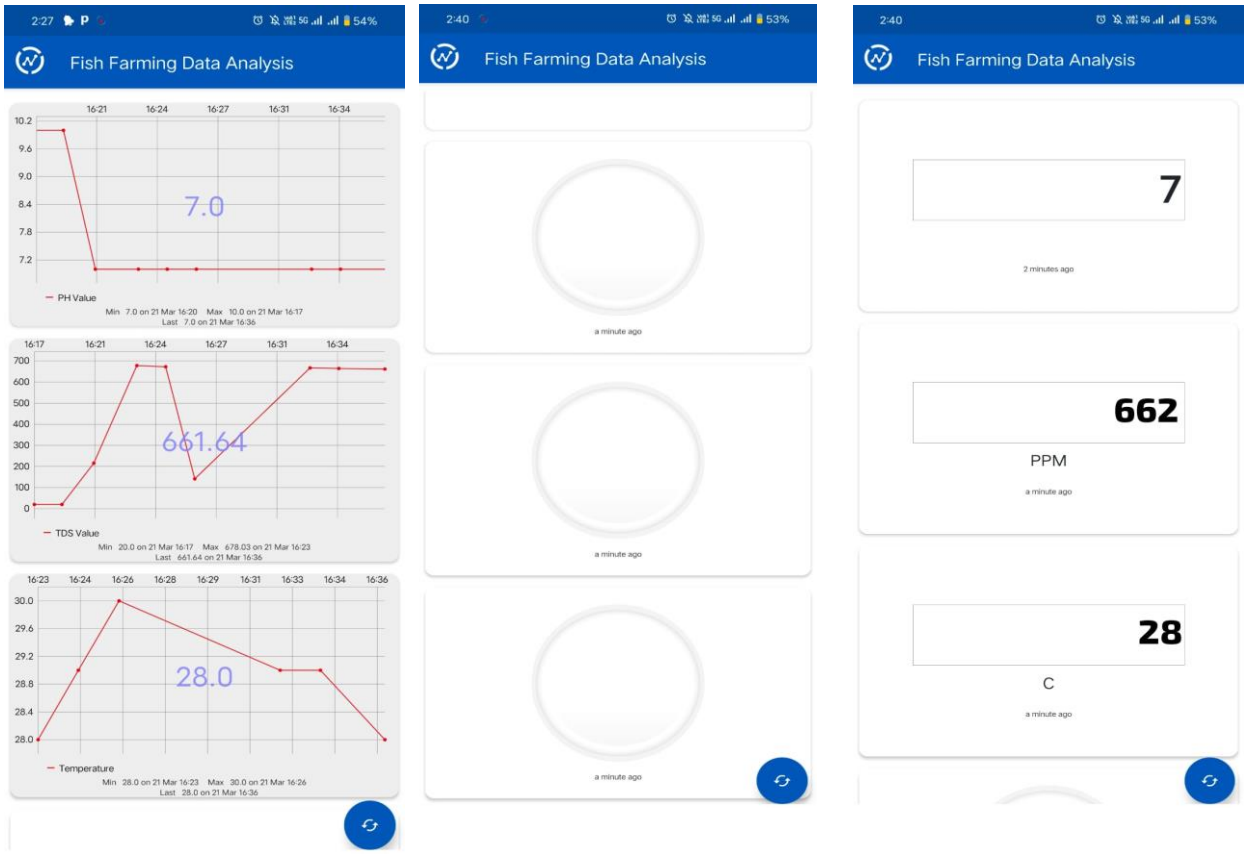


Figure 7.6 Output Waveform 1

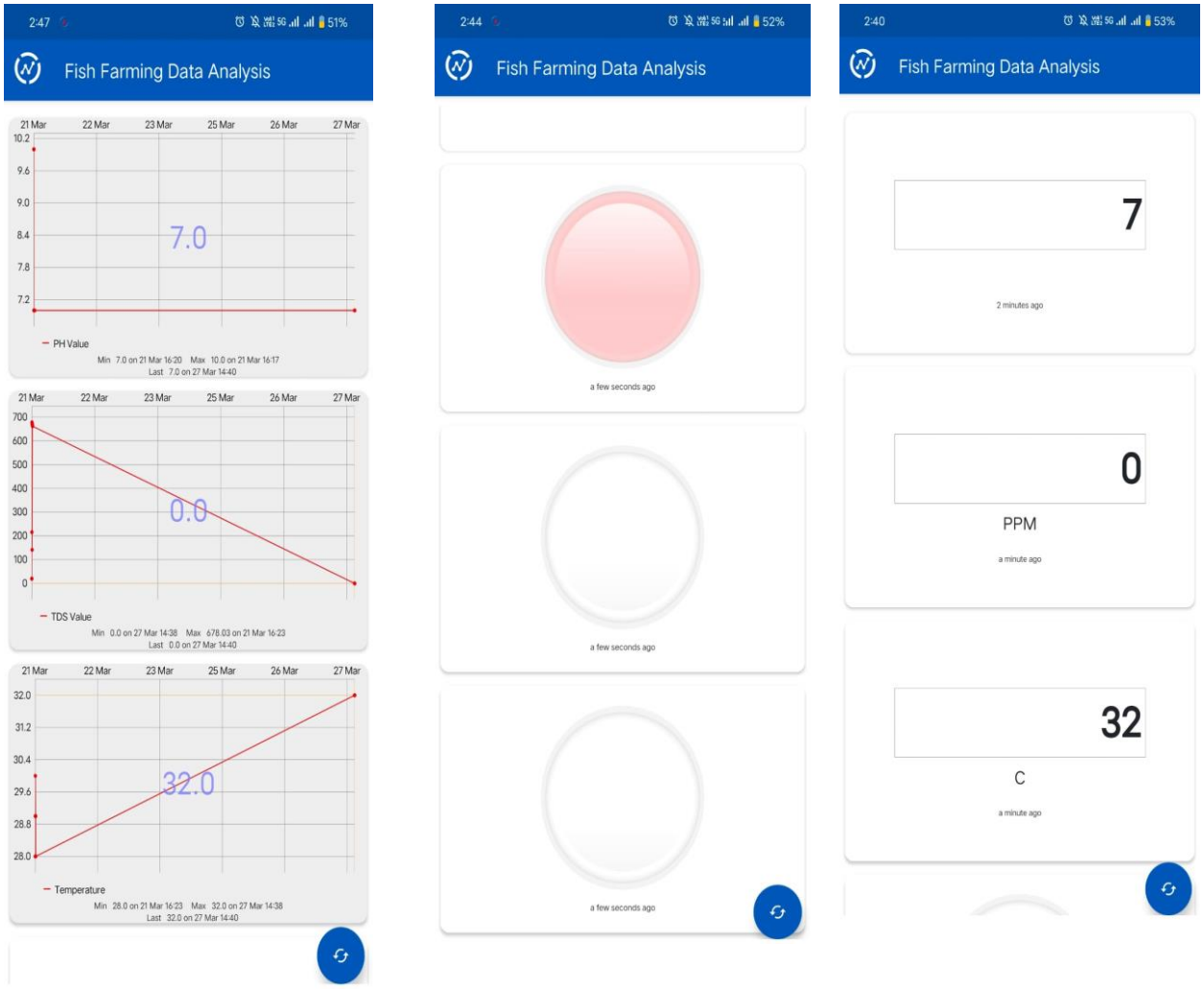
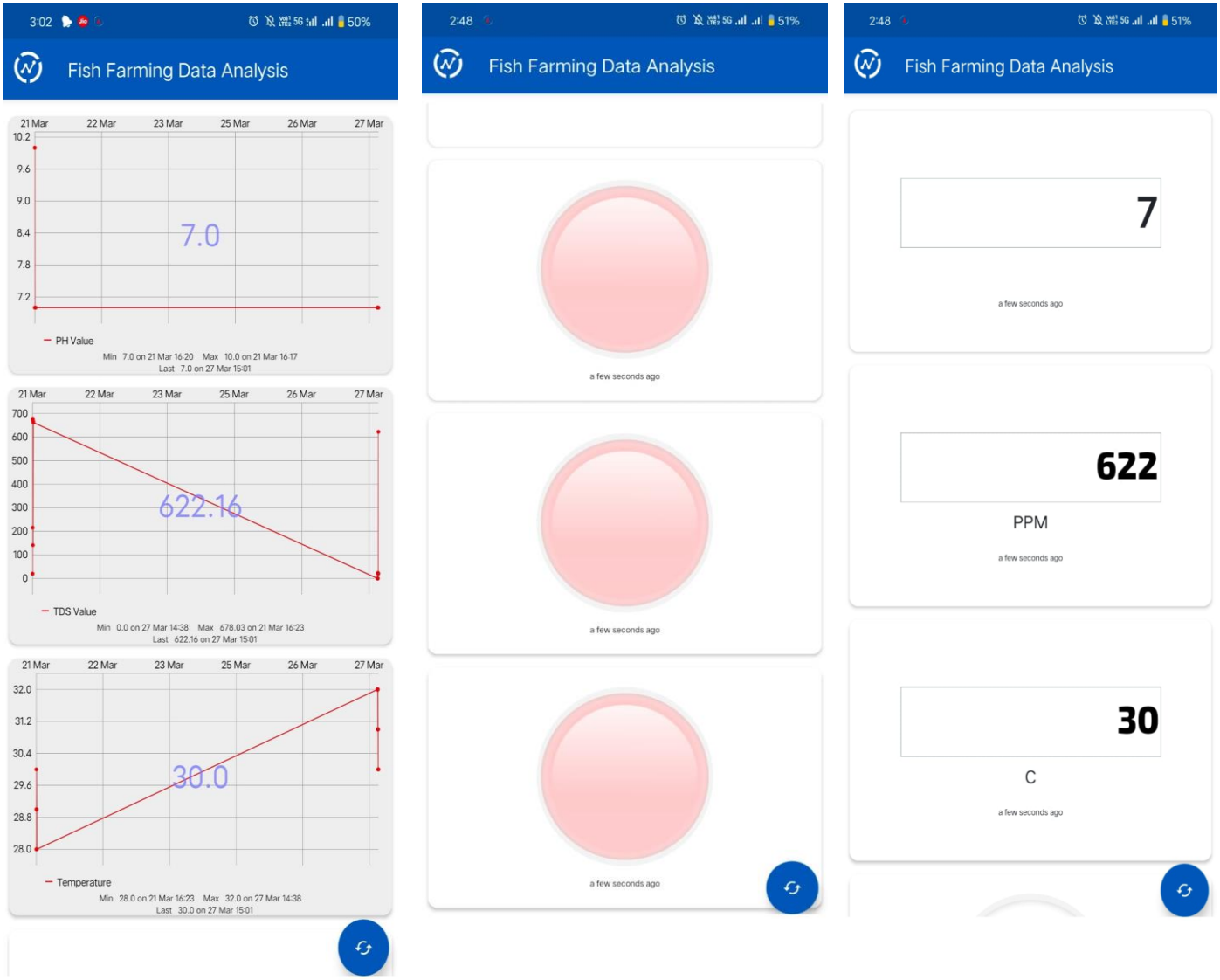


Figure 7.7 Output Waveform 2



Figures 7.8 Output Waveform 3

APPENDIX

```
#include <OneWire.h>
#include <DallasTemperature.h>
#define ONE_WIRE_BUS A4
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);
DeviceAddress insideThermometer;
#include <EEPROM.h>
#include "GravityTDS.h"
#define TdsSensorPin A6
GravityTDS gravityTds;
float tdsValue = 0;
int ph = 7;
float ammo = 1.5 ;
float oxy = 6.5 ;
#include <SoftwareSerial.h>
SoftwareSerial gprsSerial(2, 3);
#include <String.h>
int temperature;

void temper() {
  sensors.requestTemperatures();
  temperature = sensors.getTempCByIndex(0);
  Serial.print("Temp=");
  Serial.println(temperature);
  delay(1000);
}
void tdss() {
  gravityTds.update();
  tdsValue = gravityTds.getTdsValue();
  Serial.print("TDS = ");
  Serial.print(tdsValue, 0);
  Serial.println("ppm");
  delay(1000);
}
void thingspeak() {
  if (gprsSerial.available())
    Serial.write(gprsSerial.read());

  gprsSerial.println("AT");
  delay(1000);
```

```
  gprsSerial.println("AT+CPIN?");
  delay(1000);

  gprsSerial.println("AT+CREG?");
  delay(1000);

  gprsSerial.println("AT+CGATT?");
  delay(1000);

  gprsSerial.println("AT+CIPSHUT");
  delay(1000);

  gprsSerial.println("AT+CIPSTATUS");
  delay(2000);

  gprsSerial.println("AT+CIPMUX=0");
  delay(2000);

  ShowSerialData();

  gprsSerial.println("AT+CSTT=\"www\"");  //start task and setting the APN,
  delay(1000);

  ShowSerialData();

  gprsSerial.println("AT+CIICR");  //bring up wireless connection
  delay(3000);

  ShowSerialData();

  gprsSerial.println("AT+CIFSR");  //get local IP adress
  delay(2000);

  ShowSerialData();

  gprsSerial.println("AT+CIPSPRT=0");
  delay(3000);

  ShowSerialData();

  gprsSerial.println("AT+CIPSTART=\"TCP\",\"api.thingspeak.com\",\"80\"");
//start up the connection
  delay(6000);
```

```
  ShowSerialData();

  gprsSerial.println("AT+CIPSEND");  //begin send data to remote server
  delay(4000);
  ShowSerialData();
  String str = "GET
https://api.thingspeak.com/update?api_key=TQSMXKHR6SB3Y5K1&
field1=" + String(ph) + "&field2=" + String(tdsValue) + "&field3=" +
String(temperature)+ "&field4=" + String(ammo)+ "&field5=" + String(oxy);
  Serial.println(str);
  gprsSerial.println(str);  //begin send data to remote server
  delay(4000);
  ShowSerialData();
  gprsSerial.println((char)26);  //sending
  delay(5000);                   //waitting for reply, important! the time is base on the
condition of internet
  gprsSerial.println();
  ShowSerialData();
  gprsSerial.println("AT+CIPSHUT");  //close the connection
  delay(100);
  ShowSerialData();
}
void ShowSerialData() {
  while (gprsSerial.available() != 0)
    Serial.write(gprsSerial.read());
  delay(5000);
}
void setup() {
  Serial.begin(9600);
  gprsSerial.begin(9600);
  sensors.begin();
  gravityTds.setPin(TdsSensorPin);
  gravityTds.setAref(5.0);
  gravityTds.setAdcRange(1024);
  gravityTds.begin();
}
void loop() {
  temper();
  tdss();
  Serial.println("PH = 7 ");
  thingspeak();
  delay(5000);
}
```

REFERENCE

[1] C.N. Udanor, N.I. Ossai, E.O. Nweke, B.O. Ogbuokiri, A.H. Eneh, C.H. Ugwuishiwu, et al., "An internet of things labelled dataset for aquaponics fish pond water quality monitoring system", Data in Brief, vol. 43, pp. 108400, 2022.

[2] Atish Shinde, Prashant Thakare, Nikita Gilbile, Prof. M.A. Maindarkar "GPS/GSM Enabled Personal Tracker", IJSRD - International Journal for Scientific Research & Development , 2016.

[3] J. Kurnik, M. Jankovec, K.Brec, M. Topic, "Outdoor testing of PV module temperature and performance under different mounting and operational conditions," Solar Energy Materials and Solar Cells, vol. 95, no. 1, pp. 373-376, January 2011.

[4] L. Manjakkal, S. Mitra, Y. Pettilot, J. Shutler, M. Scott, M. Willander, et al., "Water Quality Monitoring using Connected Sensors Innovative Sensor Deployment and Intelligent Data Analysis", IEEE Internet of Things Journal, 2021

[5] S. U. N. Goparaju, S. S. S. Vaddhiparthy, C. Pradeep, A. Vattem and D. Gangadharan, "Design of an IoT System for Machine Learning Calibrated TDS Measurement in Smart Campus", 7th IEEE World Forum Internet Things WF-IoT 2021, pp. 877-882, 2021.

[6] KVDP Piston_vs_Plunger_pump.png, May 2010, [online] Available: Widipedia.org/wiki/piston_pump.

[7] IANG Yang, LIU Hong Yi, TIAN Honghai, HUANG Yang, " Integrated control study on testing system of solenoid valve" in Journal of NortheasternUniversity (Natural Science).Vol.31, No. 1,103-106.2019

[8] Gilbert M. Masters, "Renewable and Efficient Electric Power Systems" in , John Wiley & Sons, Inc, 2020

[9] D. Renner, P. Jansen, D. Vergossen, W. John and S. Frei, "Optimization of a 12V Dual-Battery-System for Applications in Mild-Hybrid Vehicles", AmE 2016 - Automotive meets Electronics; 7th GMM-Symposium, pp. 1-6, 2016.

[10] M. A. G. Maureira, D. Oldenhof and L. Teernstra, "ThingSpeak–an API and Web Service for the Internet of Things", World Wide Web, 2018.